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臺北市立第一女子高級中學

113 學年度科學班甄選入學

語文能力檢定試題本

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讀完本頁的說明，聽從監試委員的指示才開始作答！

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國文試題

請閱讀以下文字，按照題意要求，完成一篇作文。

從海洋到陸地，塑膠汙染的危害已進入食物鏈，成為遍及世界各地的問題，越來越多國家開始禁止塑膠袋的使用。但事實上，最早的時候，塑膠袋是為了拯救地球而發明的「環保袋」。

1959年，瑞典工程師 Sten Gustaf Thulin 發明了第一個塑膠袋。在 Thulin 看來，當時普遍使用的紙袋使森林被過度砍伐，相較之下，塑膠袋比紙袋堅固得多，加上可重複使用的優勢，理論上能解決紙袋不耐用所導致的資源耗費與環境問題。

然而這個最初想保護環境的「美意」，最終卻走向完全相反的方向。

Thulin 發明的塑膠袋後來由 Celloplast 獲得專利，沒過多久便在歐洲成為紙袋和布袋的替代品，至1979年，塑膠袋已占歐洲袋裝市場的80%。

1982年，美國最大的兩家連鎖超市 Safeway 和 Kroger 也改用塑膠袋，至2000年左右，塑膠袋幾乎已經取代全世界的紙袋。

在這樣的潮流下，一次性塑膠起飛了。現在，塑膠垃圾已經成為全球海洋面臨的最大威脅之一。根據聯合國估計，2019年塑膠袋的年產量約為一兆個，至2050年，海洋中塑膠垃圾的數量很可能超過魚類。隨著塑膠汙染日益嚴重，各國政府都在加強呼籲民眾做好回收，科學家也正在尋求更有效的解決方案。(改寫自商業週刊〈塑膠袋的誕生，竟然是為了環保？發明家之子還原發明初衷〉)

塑膠袋的出現原是科學家為拯救地球而發明，這個發明的確有一時之效，但卻帶來更深遠的危害。這個結果不禁令人疑惑：為何科學發明無法根本性解決生活上的問題？是人類能力有限，所以侷限了科學的表現？還是社會發展跟不上科學的腳步？抑或是其他原因？請根據上文事例與自己的經驗或見聞，以「科學的侷限」為題，闡述你的看法，文長 600 字為限。（占 50 分）

試題結束

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英文試題

作答提示：

請根據題組內容，在答案卷上用**英文**寫出正確、完整的答案（英翻中除外）

題組一：14 分

In 2024, NASA is going on a space adventure with the Parker Solar Probe, which is as exciting as when astronauts first landed on the moon in 1969. This special spacecraft, made by scientists at Johns Hopkins University, is going to fly into unprecedented **proximity** to the Sun—just over 6 million kilometers away. That is the closest humans have ever got to the Sun. Plus, it will fly extremely fast, about 195 kilometers every second.

However, getting close to the Sun is not easy. The Probe will face heat over 1371°C, which is really, really hot. Scientists have worked hard to make sure the spacecraft can stand the heat while gathering data about the Sun.

This mission is important because it will help us know more about the Sun. To start with, the Sun sends out solar wind, which is a group of particles that can affect the Earth a lot, like stopping our satellites and power supplies from working. The Parker Solar Probe will collect important data to understand solar wind's origin. Hopefully, the information can answer such questions as how solar wind speeds up and what the sources of high-energy solar particles are.

Besides, the Probe will look into the solar cycle. Right now, we're in a cycle that started in 2019. During this cycle, the Sun's energy has been causing different problems for our technology on the Earth, like airplanes and GPS. In 2025, the solar cycle will reach its peak, which means more solar activity. And Parker will be there watching up close the difference in the activities of the Sun, flying through the Sun's mysterious outer atmosphere. It's like Parker is surfing on the Sun's waves!

 astronaut 太空人	unprecedented 前所未有的	particle 粒子
satellite 衛星	supply 供應	peak 高峰
atmosphere 大氣層		

1. (T / F) The Parker Solar Probe mission has found the source of the Sun's particles.
2. What can be learned from the passage about the Parker Solar Probe?
(A) It will send satellites into space.
(B) It can work against the Sun's heat.
(C) It has been developed since 1969.
(D) It is made by scientists of NASA.
(E) It will go around the Sun at a record-breaking speed.
3. It can be inferred from the passage that the word **proximity** means _____.
(A) the state of being near in distance
(B) the situation of being challenged
(C) the condition of being huge in size
(D) the chance of being hit and damaged
(E) the possibility of being controlled from far away

4. Why is the 1969 moon landing mentioned in the first paragraph?
- (A) To tell where the Parker Solar Probe is going.
 - (B) To describe how the Parker Solar Probe will land.
 - (C) To report who will join in the Parker Solar Probe mission.
 - (D) To show how important the Parker Solar Probe mission is.
 - (E) To explain what dangers the Parker Solar Probe mission will face.
5. What is one of the things that the Parker Solar Probe will study?
- (A) When solar wind is created.
 - (B) What solar particles can turn into.
 - (C) How much longer the Sun can last.
 - (D) How the Sun's activity changes at the peak of a solar cycle.
 - (E) Which part of the Sun gives out the strongest solar energy.
6. What can solar wind do to the Earth?

題組二：16 分

A new study explores the interesting world of dog and cat relationships in homes where they live together. Researchers conducted a survey with people who had both dogs and cats to see how these pets got along from the owners' perspectives. They try to understand the relationships between dogs and cats better and what decides whether they can be friends or not.

According to the survey, many pet owners felt that their dog and cat lived together in peace. However, not many of them thought their pets were very close friends. In fact, cats were more likely to be threatening to dogs, but dogs seldom acted in the same way to cats. What's more, sharing food, toys, or beds did not happen often. Sometimes dogs and cats cleaned each other, but this was also rare. These findings show that while dogs and cats can live together, they might not share everything or act in a friendly way all the time.

The researchers found some factors that might make a good relationship between a dog and a cat more likely. For example, it helped if they met each other when they were young. And this was especially true for cats who were comfortable with dogs. If the cat was in the home first, it might also feel more comfortable with a dog coming later. Moreover, cats that stayed indoors showed more friendliness towards dogs than cats that went outside. **Surprisingly, whether a pet was male or female did not seem to matter much in the interaction between the two animals.**

An interesting point from the study is that it is the cat that usually decides its relationship with the dog. Cats are more likely to feel uneasy around dogs than the other way around, and they show more aggressive behavior towards dogs than dogs do towards them. The researchers think this might be because cats are not as used to being around other kinds of animals as dogs are. Another explanation may be that cats are generally smaller than dogs and they are seen by owners as more vulnerable. Therefore, owners may be more willing to accept a cat's unfriendly behavior toward a dog since a dog is unlikely to be seriously hurt by a cat. This study allows us to gain insight into the complex relationships between dogs and cats in order to keep both animals less stressed.

 survey 問卷

perspective 觀點

threatening 有威脅性的

aggressive 有攻擊性的

vulnerable 易受傷害的

1. (T / F) The study found that only a small number of cats were seriously injured by dogs.
2. What is the best title for this passage?
 - (A) What Makes a Good Pet owner?
 - (B) Understanding Behaviors of Pets under Threat
 - (C) Between Cats and Dogs: War or Peace in a House?
 - (D) Which is Really Man's Best Friend—the Dog or the Cat?
 - (E) The Roles of Pets in Keeping Peace among Family Members
3. According to the results of the survey, what could pet owners do to help their cats and dogs get along well in the house?
 - (A) Make sure they are of the same size.
 - (B) Clean both of them together more often.
 - (C) Take the dog and cat for walks separately.
 - (D) Have the dog in the house before having the cat.
 - (E) Keep both of them as pets when they are little.

4. What word in the passage is NOT used to describe a cat?

- (A) Rare.
- (B) Uneasy.
- (C) Vulnerable.
- (D) Aggressive.
- (E) Threatening.

5. Here is a description of domestication:

“Domestication occurs when humans keep animals over time to make them more useful or friendly for living together. One of signs of an animal being domesticated is its ability to live peacefully with humans and other domesticated animals.”

According to this description, which of the two animals mentioned in the passage seems more domesticated than the other?

6. Please translate the following sentence into Chinese:

Surprisingly, whether a pet was male or female did not seem to matter much in the interaction between the two animals.

題組三：20 分

Touchscreens are going hands-free with a new device that allows users to scroll through smartphones using only their tongues. MouthPad[^], a device that sits on the roof of the mouth, appeared for the first time at the Consumer Electronics Show at the beginning of 2024. It can sense tongue movements, allowing users to scroll, type, make calls and even play chess with a swipe or a click of their tongues.

The company Augmental created MouthPad[^] to be a helpful tool for the disabled, like those who lost hands or could not move. It connects to any tablet, phone or computer through Bluetooth. “It is a mouse for your mouth,” says Corbin Halliwill, a software engineer at Augmental.

MouthPad[^] is clear around the teeth, and the center is a golden touchpad, which is the contact point for the tongue. Inside there is also a force sensor that picks up left and right clicks or could be mapped to other hotkey choices. On the side, a small bump that holds the Bluetooth antenna and a wireless charging battery sticks out against the cheek. Leaving MouthPad[^] in for meals is not recommended, but it is safe to drink with it in—Halliwill says he wears it in the office while drinking water and his morning coffee.

The Augmental team has been working on its **prototype** of MouthPad[^] for about two years. The company hopes to build the technology out and introduce a new age of hands-free smartphones, bringing new possibilities for users in the coming months. Some additions may include voice and wheelchair control. The battery now lasts about five hours, but Augmental hopes to make it as long as eight hours in their next version.

MouthPad[^] interests not only the people it was made for. The company has talked with astronauts who would like to try it out at the International Space Station. **The product is expected to hit the market later this year, and more information about pre-orders can be found on their website now.**

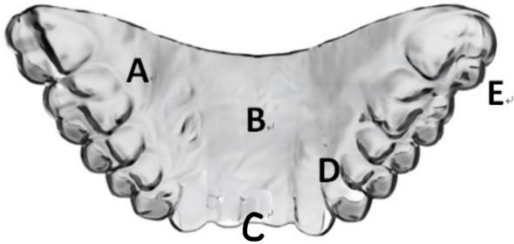
 device	裝置	scroll	滑	swipe	滑動
sensor	感應器	bump	突起物	antenna	天線
wireless	無線的	version	版本		

1. What changes may be made to improve Mouthpad[^]?
2. What does the word **prototype** mean?
 - (A) A final product that is fully developed.
 - (B) A free product given to customers to try.
 - (C) A technology that appeared in the old days.
 - (D) A cheaper version of a much more expensive product.
 - (E) The first model of a product from which future ones are developed.
3. In which paragraph can the sentence “Halliwill, along with the company’s co-founders, Corten Singer and Tomás Vega, developed the device after friends and family who were paralyzed struggled to find the right technology that met their needs” most likely appear?
 - (A) Paragraph 1.
 - (B) Paragraph 2.
 - (C) Paragraph 3.
 - (D) Paragraph 4.
 - (E) Paragraph 5.

4. What activity should users NOT do when MouthPad^ is in use?

- (A) Having Tea.
- (B) Talking on the phone.
- (C) Enjoying breakfast.
- (D) Playing board games.
- (E) Moving in a wheelchair.

5. Look at the following picture. When using MouthPad^, users can click right or left buttons by touching _____ with their tongues, and _____ is where batteries are charged.



6. Please translate the following sentence into Chinese:

The product is expected to hit the market later this year, and more information about pre-orders can be found on their website now.

試題結束

